

Energy dependence of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ S factor, and charge symmetry in the ${}^7\text{Li} + n$ and ${}^7\text{Be} + p$ systems

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Abstract

The measured energy dependence of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ S factor is fitted better by the cluster-model calculation of Descouvemont and Baye than by other models. It is suggested here that this may be connected with the cluster model predicting too high an absolute value of the S factor. Evidence for a breakdown of charge symmetry between the ${}^7\text{Li} + n$ and ${}^7\text{Be} + p$ systems is discussed and found not to be convincing. R -matrix formulae are used.

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1. Introduction

The most recent and apparently the most accurate direct measurement of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross section has been made by Junghans et al. [1–3], covering the c.m. energies $E = 116$ – 2460 keV. Their values for the astrophysical S factor, $S(E)$, are given in Table III of Ref. [2]. Junghans et al. found that their energy dependence of $S(E)$ for $E \leq 1200$ keV was better fitted by the GCM cluster-model calculation of Descouvemont and Baye [4] (DB) than by other calculations that used potential models, cluster models or hard-sphere models. A similar conclusion had earlier been reached by Jennings et al. [5], who fitted earlier data [6–8].

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To obtain their best value of $S(0)$ (or $S(20 \text{ keV})$), which is required in the solar-neutrino problem, Junghans et al. [2] fitted their data for $E \leq 362 \text{ keV}$ using the energy dependence of DB, and extrapolation gave $S(0) = 22.1 \pm 0.6 \text{ eV b}$. By fitting the same data with other calculations, and by including data from other recent direct measurements [7–11], they obtained

$$S(0) = 21.4 \pm 0.5(\text{expt}) \pm 0.6(\text{theor}) \text{ eV b.} \quad (1)$$

Because of the large theoretical uncertainty, they concluded that additional theoretical work to reduce the extrapolation uncertainty would be very valuable.

Junghans et al. [2,3] also compared the $S(E)$ energy dependence obtained from these direct measurements with that inferred from four recent Coulomb dissociation (CD) measurements [12–15]. While the direct measurements are well fitted by the DB energy dependence, the CD measurements are not; they have a significantly steeper slope and are in agreement with the energy dependence given by potential models [13,15].

A solution of this discrepancy between the direct and CD slopes has recently been proposed by Esbensen et al. [16]. They claim that the conventional first-order analyses of the CD data were deficient, and that more-consistent slopes are obtained by including effects of higher order in the Coulomb and nuclear fields. We assume that this explains the CD results, and so concentrate on the $S(E)$ energy dependence from the direct measurements, and consequently on that of DB. We attempt to understand the difference between the energy dependence given by DB and by other calculations.

2. DB cluster-model calculation

DB made a microscopic study of the ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ and ${}^7\text{Be}(p, \gamma){}^8\text{B}$ mirror reactions in a multiconfiguration three-cluster GCM approach. In an earlier calculation [17], Descouvemont and Baye used a similar model with $(\alpha + t)_{L=1} + n$ or $(\alpha + {}^3\text{He})_{L=1} + p$ clusters, and the Volkov interaction V2. DB [4] included additional configurations and also used V1, V3 and V4 interactions. Junghans et al. [1–3] used the DB energy dependence obtained with the V2 interaction, for which DB say that their $S(E)$ is identical to that obtained in their earlier calculation [17], and that their ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ cross section is very similar to that found in [17] (except near the 3^+ resonance, which was missing in [17]). Thus we also include some of the information given in [17] in discussing the DB results (which we restrict to those with the V2 interaction).

Is the DB model inherently better than others? If so, one would expect its predictions for relevant quantities in addition to the $S(E)$ energy dependence to be in good agreement with experiment. We note immediately that the absolute value of $S(E)$ predicted by DB is much larger than experimental values (note the scaling factor $C_1 \approx 0.7$ in Eq. (11) of Ref. [2]). DB give $S(0) = 30 \text{ eV b}$, to be compared with Eq. (1).

DB say that “The ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ reaction ... provides a valuable test of models dedicated to the mirror ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction”. DB do not give their ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ cross section for thermal neutrons, $\sigma_{n\gamma}(E_{\text{th}})$, for comparison with the experimental value of $45.4 \pm 3 \text{ mb}$ [18,19]. DBs cross section calculated at $E = 25 \text{ keV}$ is $54.9 \mu\text{b}$; with a $1/v$ dependence, this corresponds to a thermal value of 58.7 mb (54.8 mb in [17]). DB do not mention the ${}^7\text{Li} + n$ scattering lengths a_s (s is the channel spin). Values are given in Ref. [17]: $a_1 = -0.72 \text{ fm}$, $a_2 = -3.26 \text{ fm}$. These may be compared with experimental values: $a_1 = 0.87 \pm 0.07 \text{ fm}$, $a_2 = -3.63 \pm 0.05 \text{ fm}$ [20]. Ref. [17] also gives their values of the branching ratio $\mathcal{R}$ to the first excited state of ${}^8\text{Li}$ as 0.111 (experimental value 0.106 ± 0.010 [18]), and the fraction W_+ of the thermal-neutron capture that proceeds via the initial channel spin $s = 2$ as 0.79 (experimental value $\gtrsim 0.86$ [21]). The values

of $\mathcal{R}$ and W_+ depend on the structure of the 1^+ state; the DB calculation suggests a strong dominance of channel spin 1 ($(s=2)/(s=1) = 1/3.5 = 0.29$) [22], and this is supported by the $^7\text{Be} + p$ elastic-scattering data of Angulo et al. [22], who obtained a much better fit with $s = 1$ for the 1^+ level than with $s = 2$. An analysis of the data including channel-spin mixing (which was neglected by Angulo et al.) gives a channel-spin ratio $(s=2)/(s=1)$ of about 0.06. On the other hand, shell-model calculations (see [23]) predict $s = 2$ dominance, and the 17.64 MeV 1^+ level of ^8Be , which is mainly the $T = 1$ analog of the 1^+ state of ^8B , has an experimental channel-spin ratio of 3.2 ± 0.5 [24]; after allowing for isospin mixing, the ratio for the pure $T = 1$ state can, however, be appreciably smaller than this (see Table 3 of Ref. [25]).

It has been argued [26] that the results of $^7\text{Li}(n, \gamma)^8\text{Li}$ calculations need not indicate the quality of the calculated $^7\text{Be}(p, \gamma)^8\text{B}$ cross section, because the former are sensitive to internal contributions while the latter is not. A good model should, however, describe the internal contributions correctly and consequently should fit $^7\text{Li}(n, \gamma)^8\text{Li}$ data.

Standard potential-model calculations [23,27] overestimate the $^7\text{Li}(n, \gamma)^8\text{Li}$ cross section. Brown et al. [27] have suggested that the calculated $^7\text{Li}(n, \gamma)^8\text{Li}$ cross section could be reduced to agree with experiment without affecting the $^7\text{Be}(p, \gamma)^8\text{B}$ cross section, by replacing the potential-model scattering wave function in the internal region by that obtained in a cluster-model calculation [28]. They still use the potential-model bound-state wave function; the argument would be more convincing if the initial and final states came from the same model. They imply that the cluster-model scattering wave function reproduces the experimental $^7\text{Li} + n$ scattering lengths, but this seems doubtful (the $^7\text{Li} + n$ system is not mentioned in Ref. [28]).

All the models mentioned above [4,17,23,26,27] assume charge symmetry between the mirror systems $^7\text{Li} + n$ and $^7\text{Be} + p$. Some evidence for the breakdown of charge symmetry has been provided by Angulo et al. [22], who studied $^7\text{Be} + p$ elastic scattering. Their value of the scattering length for channel spin 1, $a_1 = 25 \pm 9$ fm, is not consistent with the value of 2 to 3 fm that they expect from charge symmetry and the experimental value of a_1 for $^7\text{Li} + n$. This is discussed further in Section 4.

According to Ref. [28], the DB model gives ^7Be unbound relative to the $^4\text{He} + ^3\text{He}$ threshold; experimentally it is bound by 1.59 MeV. Ref. [28] also says that the V2 interaction underbinds the deuteron by 1.6 MeV, while binding the (experimentally unbound) singlet dinucleon states. Recently the DB calculation for $^7\text{Be}(p, \gamma)^8\text{B}$ has been updated and improved by Descouvemont [29], who allowed the ^7Be cluster to be distorted, and used the Minnesota interaction (MN) in addition to the V2 interaction. The distortion had little effect on $S(0)$; with the V2 interaction the new calculation gives $S(0) = 29.45$ eV b (compared with 30 eV b in DB). With the MN interaction, $S(0) = 24.69$ eV b.

Other quantities of significance in the resonant contributions to the $^7\text{Li}(n, \gamma)^8\text{Li}$ and $^7\text{Be}(p, \gamma)^8\text{B}$ cross sections (but not directly concerning $S(0)$), are the energy, width and radiative width of the 3^+ level of ^8Li and of the 1^+ and 3^+ levels of ^8B . DB give for these (with experimental values [24] given in square brackets) $E(^8\text{Li}, 3^+) \approx 0.6$ MeV [0.22 MeV], $\Gamma(^8\text{Li}, 3^+) = 44$ keV [33 ± 6 keV], $\Gamma_\gamma(^8\text{Li}, 3^+ \rightarrow 2^+) = 0.19$ W.u. [0.29 ± 0.13 W.u.], $E(^8\text{B}, 1^+) \approx 0.2$ MeV [0.63 MeV], $\Gamma(^8\text{B}, 1^+) = 59$ keV [35.6 ± 0.6 keV], $\Gamma_\gamma(^8\text{B}, 1^+ \rightarrow 2^+) = 3.5$ W.u. [2.63 ± 0.12 W.u.], $E(^8\text{B}, 3^+) \approx 2.8$ MeV [2.18 ± 0.02 MeV], $\Gamma(^8\text{B}, 3^+) = 529$ keV [350 ± 30 keV], $\Gamma_\gamma(^8\text{B}, 3^+ \rightarrow 2^+) = 0.16$ W.u. [0.38 ± 0.19 W.u.].

DB comment briefly on a comparison of their results with those of Knox et al. [30]. Knox et al. fitted a large body of $^7\text{Li} + n$ data with multilevel, multichannel R -matrix formulae, taking initial values of the level energies and reduced widths from a shell-model calculation, and needing to change only a few of these in order to fit the data. In particular, Knox et al. found 1^- levels of

^8Li at $E = 1.44$ and 3.58 MeV, and a 2^- level at 1.18 MeV, with appreciable s -wave reduced widths for the $n + ^7\text{Li}(\text{g.s.})$ channel (and for the $n + ^7\text{Li}(1/2^-)$ channel for the 1^- levels). These levels could give significant s -wave E1 contributions to the $^7\text{Li}(n, \gamma)^8\text{Li}$ cross section. DB say that their model suggests broad ($\Gamma_n > 1$ MeV) 1^- and 2^- resonances near 1 MeV.

These comparisons show that there are discrepancies between DB predictions and experimental values that are often many times the experimental uncertainties. This is perhaps not unexpected, as the cluster-model calculation contains no adjustable parameters once the NN interaction is chosen. It does suggest, however, that there is no *a priori* justification for assuming that the DB $S(E)$ energy dependence should be better than others.

We now try to understand why the DB energy dependence for $S(E)$ actually gives such good agreement with experiment. For this purpose, we use the R -matrix model [31], in which it is possible to vary the properties of the levels of ^8B that could be contributing significantly.

3. R-matrix model calculation

The R -matrix parameters include channel radii, eigenenergies, spectroscopic amplitudes and photon reduced-width amplitudes. There are many sets of parameter values that fit the DB $^7\text{Li} + n$ data. Of these we choose the one that best fits the DB values of the $^7\text{Be}(p, \gamma)^8\text{B}$ S factor (Section 3.1). We then consider the changes in these parameter values required to fit the experimental data for $^7\text{Li} + n$ and their effect on the $^7\text{Be}(p, \gamma)^8\text{B}$ values. This leads to a correlation between the properties of the $^7\text{Be}(p, \gamma)^8\text{B}$ S factor at low and at higher energies (Section 3.2). We then discuss the relationship between the shape of the $^7\text{Be}(p, \gamma)^8\text{B}$ S factor and the value of $S(0)$ (Section 3.3).

To quantify the different slopes of $S(E)$ shown by the direct $^7\text{Be}(p, \gamma)^8\text{B}$ measurements and the CD measurements, Junghans et al. [2] introduced a slope parameter b , defined by $S(E) = a(1 + bE)$ and determined by fitting data in the range $E \leq 425$ keV and $830 \leq E \leq 1300$ keV. For the direct measurements they find $b = 0.311 \pm 0.014$ MeV $^{-1}$ and for the CD measurements $b = 0.55 \pm 0.08$ MeV $^{-1}$. For the DB curve given in Fig. 18 of Ref. [2], we estimate $b \approx 0.28$ MeV $^{-1}$ (by fitting $S(E)$ values at $E = 0.3$ and 1.2 MeV).

It is, however, the behaviour of $S(E)$ at lower energies ($E \lesssim 100$ keV) that is more important in extrapolating the values measured at $E > 100$ keV to obtain $S(0)$ [or $S(20$ keV)]. Jennings et al. [5] find that $S(0)/S(100$ keV) varies by nearly 5% among the various models they consider (their Table IV). We therefore are interested in the slope of $S(E)$ at $E = 0$, represented by $(d \ln S/dE)_{E=0}$ and denoted by s_1 in Ref. [32]. It is of interest to see if there is any correlation between the values of b and s_1 .

3.1. R-matrix representation of DB calculation

Firstly, we try to find R -matrix parameter values that reproduce the results of the DB calculation. The $^7\text{Li} + n$ values that we fit are (with experimental values given in square brackets):

$$\begin{array}{lll} a_1 \text{ (fm):} & -0.72 & [0.87 \pm 0.07], \\ a_2 \text{ (fm):} & -3.26 & [-3.63 \pm 0.05], \\ \sigma_{n\gamma}(E_{\text{th}}) \text{ (mb):} & 58.7 & [45.4 \pm 3.0], \\ \mathcal{R}: & 0.111 & [0.106 \pm 0.010], \\ W_+: & 0.79 & [\gtrsim 0.86]. \end{array} \quad (2)$$

We also fit the width of the 1^+ level of ^8B :

$$\Gamma(^8\text{B}, 1^+) \text{ (keV):} \quad 59 \quad [35.6 \pm 0.6]. \quad (3)$$

As before [31], we do the calculations for the channel radius $a = 4, 5$ and 6 fm, using a four-level, four-channel R -matrix approximation. The boundary condition parameters are chosen as in Ref. [31]. Fitting the scattering lengths a_s determines the zeros of the 1^- and 2^- energy scales. The remaining four values in (2), (3) are fitted by adjusting parameters to which these quantities are most sensitive. There are eight of these parameters (four spectroscopic amplitudes for the 1^+ and 2^+ levels, and four internal contributions to the photon reduced-width amplitudes). We vary at most four parameters at a time. Of the 21 modified sets of parameter values that give an exact fit (for $a = 4.0$ fm), we choose the set that predicts $S(0)$ and b closest to the DB values of 30 eV b and 0.28 MeV $^{-1}$; this set gives $S(0) = 31.2$ eV b and $b = 0.24$ MeV $^{-1}$ ($S(0)$ and s_1 are obtained by fitting $S(E)$ values calculated at $E = 10, 20$ and 30 keV using the formula (11) of Jennings et al. [5], and b is calculated by fitting $S(E)$ values at $E = 0.3$ and 1.2 MeV). For this set, $S(2^+) = 2.40$, which is much greater than the experimental value (0.87 from Ref. [31]); however, the value $S(1^+) = 1.22$ required to fit the DB width (3) is also much greater than the experimental value (0.48 from Ref. [31]), so that the ratios $S(2^+)/S(1^+)$ roughly agree. We also find $s_1 = -1.87$ MeV $^{-1}$. A check on how well this set resembles the DB calculation comes from the d -wave, $s = 2$ contribution to $S(E)$, given in Ref. [17] as 6% at 20 keV, 12% at 100 keV and 48% at 1 MeV. For this modified set, we find 7%, 12% and 47%, respectively, in good agreement. A further check comes from the values of the 1^+ spectroscopic factors; for the original set, which are shell-model values, the channel spin $s = 2$ dominates, with a channel-spin ratio ($s = 2$)/($s = 1$) of 1.85 . For the modified set, the channel-spin ratio is 0.32 , in good agreement with the DB value of 0.29 .

3.2. Relationship between b and s_1

Taking this modified set as a standard set giving a reasonable representation of the DB values of $S(E)$, in addition to the DB values given in (2), (3), we see how the output values $S(2^+)$, $S(0)$, s_1 and b change as the input values in (2), (3) are changed, one at a time. In some cases we change the DB values to the experimental values, also given in (2), (3), but in other cases this does not allow an exact fit of the data by modification of the same four parameters, and we consider smaller changes. The results of this procedure are given in Table 1. It is seen that these changes produce appreciable changes in the values of b (as well as $S(2^+)$ and $S(0)$). It seems that the DB values given in (2), (3) are too far from the experimental values for one to argue that removal of the discrepancies (if that were possible) would not change appreciably the DB

Table 1
Effect of changes to the input DB data ($a = 4.0$ fm)

Quantity changed	Change in value	$S(2^+)$	$S(0)$ (eV b)	s_1 (MeV $^{-1}$)	b (MeV $^{-1}$)
Standard set					
		2.40	31.2	-1.873	0.242
Changed sets					
a_1 (fm)	$-0.72 \rightarrow -0.10$	0.92	17.1	-1.699	0.767
a_2 (fm)	$-3.26 \rightarrow -3.63$	2.40	31.2	-1.882	0.218
$\sigma_{n\gamma}(E_{th})$ (mb)	$58.7 \rightarrow 55.0$	3.30	36.3	-1.922	0.134
$\mathcal{R}$	$0.111 \rightarrow 0.106$	2.40	31.2	-1.871	0.244
W_+	$0.79 \rightarrow 0.86$	13.41	54.6	-2.004	-0.025
$S(1^+)$	$1.22 \rightarrow 0.90$	1.08	19.2	-1.737	0.628

energy dependence of $S(E)$. So far we have used the results for $a = 4.0$ fm. For $a = 6.0$ fm, the modified fits all give $S(0)$ values near 20 or 40 eV b, far from the DB value of 30 eV b. For $a = 5.0$ fm, many of the modified sets give $S(0)$ and b values close to the DB values, and the changes as given in Table 1 have little effect on the values of $S(2^+)$, $S(0)$, s_1 and b ; the values of s_1 are, however, quite different from those for $a = 4.0$ and 6.0 fm (see Fig. 1). A possible reason for this is given in the next section.

Fig. 1 shows the relation between the values of s_1 and b obtained from all the 21 modified sets (for $a = 4.0$ fm) and from the fits listed in Table 1, in addition to similar fits made to the experimental data given in (2) (plus $S(1^+) = 0.48$, $S(2^+) = 0.87$). Corresponding values are also given for fits with $a = 5.0$ and 6.0 fm. The GCM points in Fig. 1 are discussed later. For the $a = 4.0$ fm fits, there is a strong correlation between s_1 and b , which is independent of the values of many other quantities. This suggests that the changes in s_1 and b are due to the same component of the S factor (same J_i , s and l). As s_1 is a zero-energy quantity, the different values of s_1 (and therefore of b) must be due to differences in the s -wave component of $S(E)$. This is because the d -wave contribution at zero energy is small and well-determined at about 6% [17, 23, 29, 32], and the value of s_1 calculated for the d -wave component alone varies little for the different fits represented in Fig. 1. The s -wave $s = 1$ component is small and its values of s_1 vary little, but the s -wave $s = 2$ component is dominant and its s_1 values vary considerably (from -2.42 to -2.75 for the $a = 4.0$ fm points in Fig. 1). Thus the s -wave $s = 2$ component seems to account for most of the changes of s_1 , and therefore of b , shown in Fig. 1. Descouvemont [29] also reached this conclusion, by comparing the components of $S(E)$ for the GCM cluster calculation with those from a potential-model calculation. Snover et al. [33] implicitly found a correlation between s_1 and b , while investigating the extrapolation uncertainty in $S(0)$ (see Eq. (1)). They allowed the ratio of the s -wave and d -wave contributions to $S(0)$ given by several theories to vary, and chose the ratio in each case to best fit the experimental $S(E)$ of Ref. [2] over

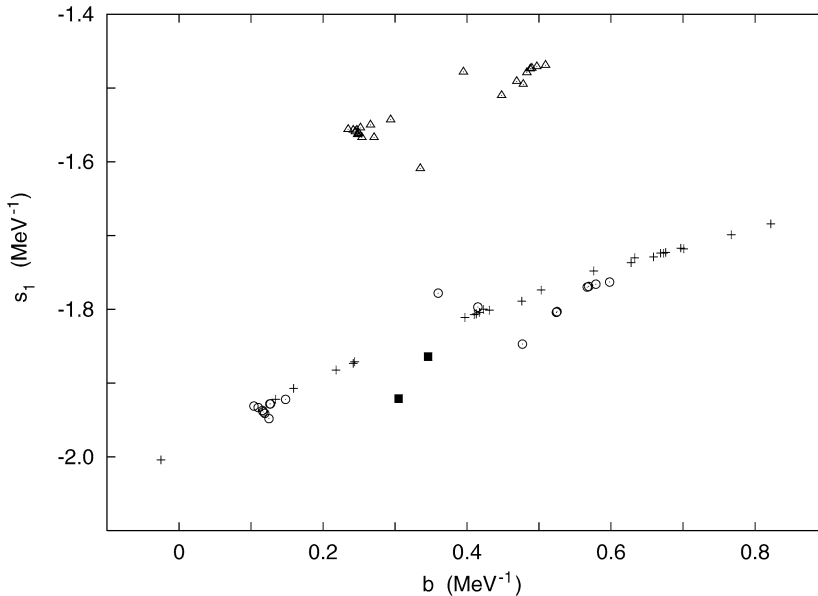


Fig. 1. Relation between s_1 and b values obtained from R -matrix fits to DB data and to experimental data, for channel radius $a = 4.0$ fm (pluses), 5.0 fm (triangles) and 6.0 fm (circles), and from GCM calculations [29] (solid squares).

a wide energy range ($E \leq 1200$ keV), so essentially fitting the DB value of b . They then found that the spread of $S(0)$ values obtained by normalizing to the low-energy data [2] ($E \leq 425$ keV) was significantly reduced; i.e., the value of s_1 was better determined. Such variations in the s/d ratio are not, however, very realistic, as all theories give essentially the same ratio at zero energy (the zero-energy d -state contribution lies between 6.04% and 6.25% for all the points in Fig. 1; Baye [32] gives 6.14%, neglecting scattering-length effects).

3.3. Relationship between b and $S(0)$

The values of b are also correlated with those of $S(2^+)$ and $S(0)$; Fig. 2 shows this for $S(0)$ for $a = 4.0, 5.0$ and 6.0 fm. Small values of b are associated with large values of $S(2^+)$ or of $S(0)$. We therefore explore the consequences of increasing the value of $S(2^+)$ or of $S(0)$, starting from parameter values that fit the experimental data given in (2). As we have attributed the changes in b to changes in the s -wave $s = 2$ component, we look at the relative sizes and energy dependences of the various terms contributing to this component. The collision matrix element for the 2^- levels is given by the formula (6) in Ref. [31] (with $J_i = 2$, $s_e = 2$, $l_i = 0$, $J_f = 2$). There are three contributions: a resonant internal term, a resonant channel term and a nonresonant channel term (often referred to as the hard-sphere capture amplitude). For simplicity, we replace the four-level, four-channel 2^- contribution by a one-level, one-channel approximation, as already justified and used in Ref. [34]. Then the summation over λ and μ in the formula (6) reduces to a single term, as does the summation over channels c .

For $a = 4.0$ fm, the parameter values that fit the experimental data in (2) (plus $S(2^+) = 0.87$) give $S(0) = 16.3$ eV b and $b = 0.54$ MeV $^{-1}$. Over the energy range of interest, say $E \lesssim 1.5$ MeV, the internal contribution is small compared with the resonant channel term. The nonresonant channel contribution (within the square brackets in the formula (6) in [31]) decreases as E in-

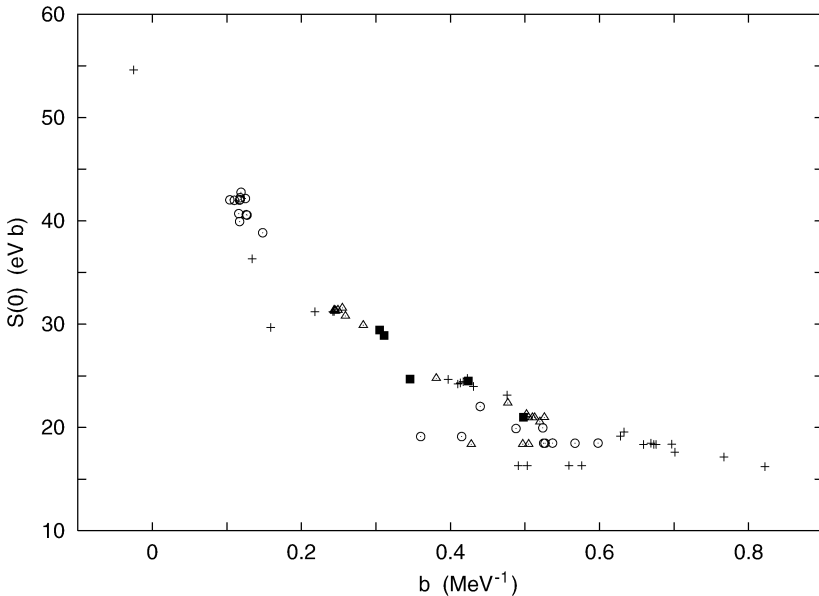


Fig. 2. Relation between $S(0)$ and b values obtained from R -matrix fits to DB data and to experimental data, for channel radius $a = 4.0$ fm (pluses), 5.0 fm (triangles) and 6.0 fm (circles), and from GCM calculations [29,36] (solid squares).

creases, but is larger than the total resonant contribution over the whole energy range. The real part of the resonant contribution has the same sign as the nonresonant contribution throughout the energy range. Fitting the DB data (2), (3) gives $S(2^+) = 2.46$, $S(0) = 31.6$ eV b, and $b = 0.22$ MeV⁻¹. The channel terms, both resonant and nonresonant, increase as they are approximately proportional to $\sqrt{S(2^+)}$. In order to fit the ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ data, the internal amplitude also increases in magnitude, to be comparable with but of opposite sign to the resonant channel amplitude. For ${}^7\text{Be}(p, \gamma){}^8\text{B}$, the magnitude of the total resonant contribution is then of the same order as for $S(2^+) = 0.87$, and it is therefore small compared with the nonresonant contribution. The phase of the total resonant contribution is, however, considerably different from that for $S(2^+) = 0.87$, with the real part of the resonant term changing from being of the same sign as the nonresonant term for $E < 1$ MeV to being of opposite sign for $E > 1$ MeV. Consequently, as E increases, the total 2^- contribution to the S factor decreases more rapidly for the larger value of $S(2^+)$. The total S factor then increases more slowly for larger $S(2^+)$, giving the smaller value of b .

It seems that the better fit of DB to the experimental energy dependence of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ S factor may be due to a large effective value of $S(2^+)$ (coming from their large value of $S(0)$). Descouvemont [29] gives a different explanation. He finds that the energy dependence $S(E)/S(0)$ given by his GCM calculation is changed if the ${}^7\text{Be}$ deformation is changed, and approaches that of a potential model [35] as the deformation approaches zero. The two explanations seem, however, to be related. The GCM calculation [29] finds that the magnitude of $S(0)$ decreases as the ${}^7\text{Be}$ deformation decreases [36]. Also b increases as the deformation decreases [29]. The relation between $S(0)$ and b for three values of the deformation ($R_2 = 0, 2.4$ and 3.7 fm), and for the full calculations with the MN and V2 interactions, is shown in Fig. 2 (solid squares). There appears to be consistency between the GCM and R -matrix results.

4. Charge symmetry

In Section 2, there was mention of some evidence from ${}^7\text{Be} + p$ elastic scattering [22] that charge symmetry may not be valid for the ${}^7\text{Li} + n$ and ${}^7\text{Be} + p$ systems. We here use R -matrix calculations to test if the ${}^7\text{Be} + p$ data [22] can be fitted when charge symmetry is assumed. Firstly we use R -matrix parameter values from Section 3, which fit ${}^7\text{Li} + n$ scattering lengths (and other ${}^7\text{Li} + n$ data) to calculate ${}^7\text{Be} + p$ scattering lengths and effective ranges, assuming charge symmetry. This leads to doubts about some of the procedures used in the analysis in Ref. [22]. We then directly fit the ${}^7\text{Be} + p$ elastic-scattering data [22] using sets of R -matrix parameter values that also fit the ${}^7\text{Li} + n$ data (therefore assuming charge symmetry), and find acceptable fits for some sets (these are in group A in Tables 2 and 3). Improved fits to the ${}^7\text{Be} + p$ data are possible by allowing a departure from charge symmetry (group B in Tables 2 and 3).

Angulo et al. [22] analysed their ${}^7\text{Be} + p$ elastic-scattering measurements assuming that the s -wave phase shifts could be represented by effective-range approximations, and determined the scattering lengths a_s for channel spins 1 and 2 to be $a_1 = 25 \pm 9$ fm and $a_2 = -7 \pm 3$ fm. From the experimental values of a_s for ${}^7\text{Li} + n$ and the assumption of charge symmetry, they estimated for ${}^7\text{Be} + p$ the values $a_1 = 2$ to 3 fm and $a_2 = -7$ to -11 fm. While there is consistency for a_2 , there is a clear inconsistency for a_1 , suggesting a breakdown of charge symmetry.

To relate the values of a_s for ${}^7\text{Li} + n$ and ${}^7\text{Be} + p$, Angulo et al. [22] used a potential model that embodies charge symmetry; they found that, for any given value of a_s for ${}^7\text{Li} + n$, the predicted value of a_s for ${}^7\text{Be} + p$ is well determined, as are the values of the effective ranges r_{es} for both

Table 2

Values of a_s and r_{es} for ${}^7\text{Li} + n$ and ${}^7\text{Be} + p$ from R -matrix formulae

Group	${}^7\text{Li} + n$					${}^7\text{Be} + p$			
	a (fm)	a_1 (fm)	r_{e1} (fm)	a_2 (fm)	r_{e2} (fm)	a_1 (fm)	r_{e1} (fm)	a_2 (fm)	r_{e2} (fm)
A	4.0	0.92 ^a	−21.7	−3.63 ^a	2.33	3.33	−5.48	−4.34	−1.37
	4.5	0.93 ^a	−32.8	−3.63 ^a	1.30	5.15	−3.19	−2.32	−20.0
	5.0	0.95 ^a	−47.3	−3.63 ^a	−0.57	7.84	−1.50	0.90	−319.2
	5.3	0.96 ^a	−57.8	−3.63 ^a	−2.01	9.90	−0.81	3.43	−24.8
	5.4	0.96 ^a	−61.6	−3.67 ^a	−2.24	10.7	−0.59	4.25	−16.6
	5.5	0.96 ^a	−65.6	−3.63 ^a	−3.10	11.5	−0.41	5.38	−10.7
	6.0	0.96 ^a	−88.9	−3.63 ^a	−6.05	16.0	0.30	11.2	−2.25
B	6.5	0.91 ^a	−126.1	−3.63 ^a	−9.23	21.4	0.76	18.5	−0.11
	4.0	3.72	2.46	−4.30	1.95	18.8	1.60	−4.43	−1.90
	4.5	3.53	2.01	−3.95	3.04	18.1	1.48	−3.41	−6.43
	5.0	4.05	2.36	−10.1	4.80	24.8	1.64	−5.06	−3.73
	5.3	3.56	0.34	−8.56	5.94	24.6	1.49	−4.30	−6.41
	5.4	3.51	−0.31	−13.4	5.29	25.6	1.49	−3.86	−12.2
	5.5	3.10	−3.08	−13.7	5.10	24.3	1.38	−2.59	−35.1
	6.0	1.51	−26.8	−22.4	4.37	16.9	0.55	4.18	−30.9
	6.5	−2.31	−18.1	−16.8	5.75	14.9	−0.64	7.64	−9.54

^a Fitted to experimental value.

${}^7\text{Li} + n$ and ${}^7\text{Be} + p$. The potential model implies that any 1^- or 2^- level of ${}^8\text{B}$ contributing to the ${}^7\text{Be} + p$ elastic scattering is a single-particle level (spectroscopic factor $\mathcal{S} = 1$ for the ${}^7\text{Be}$ ground-state channel). This may be a reasonable assumption for channel spin 2, as shell-model calculations [30,37] predict a single low-lying 2^- level of ${}^8\text{Li}$ or ${}^8\text{B}$, with a large value of $\mathcal{S}$ (≈ 0.7 – 0.8). For $s = 1$, however, the shell-model calculations give two low-lying 1^- levels, each with $\mathcal{S} \approx 0.4$ (and with similar values of $\mathcal{S}$ for the first excited state channel). Thus it is not clear that the value of a_1 for ${}^7\text{Be} + p$ given by the potential model can be trusted.

In Section 3, R -matrix formulae are used to connect parameter values for the 1^- and 2^- levels with the scattering lengths a_1 and a_2 for ${}^7\text{Li} + n$. We now use the simple one-level, one-channel model of Ref. [34] for the 2^- levels (and a two-level, two-channel approximation for the 1^- levels), with the same values of the excitation energies and reduced widths as in Section 3, to calculate a_s for ${}^7\text{Be} + p$, as well as the values of r_{es} for both ${}^7\text{Li} + n$ and ${}^7\text{Be} + p$. Results are given in Table 2 (group A). Because some quantities are sensitive to the value of the channel radius a , Table 2 includes results for additional values of a .

From their potential model, Angulo et al. [22] found $a_s({}^7\text{Be} + p)/a_s({}^7\text{Li} + n) \approx 2$ – 3 . The values in Table 2 show no such relationship. The potential model [22] gives $r_{es} > 0$ for all values of a_s . In the R -matrix calculation, r_{es} can be negative, and it can attain large magnitudes (positive or negative). In the simple case of a single s -wave level and a single neutron channel, formulae for a_s and r_{es} have been given in terms of the resonance energy E_r and reduced width γ^2 (Eqs. (19) and (20) in Ref. [38]). From these, it is seen that $a_s = 0$ for $E_r = \gamma^2$. As $E_r \rightarrow \gamma^2$, one finds $r_{es} \rightarrow \pm\infty$ (for $\gamma^2 \geq 3\hbar^2/2\mu a^2$). For $E_r \ll \gamma^2$, one has $r_{es} > 0$; this seems to correspond approximately to the potential-model description (with $\mathcal{S} = 1$). For smaller $\mathcal{S}$ (smaller γ^2), r_{es} can become negative.

Angulo et al. [22] only considered values of r_{es} for ${}^7\text{Be} + p$ in the range 1–5 fm (earlier Baye [32] had also used values of r_{es} from 0 to 5 fm, considering the latter as a large value). It seems

that the effective range was expected to be of the order of the actual range of the nuclear potential. This is the case for nucleon–nucleon scattering, for which effective-range expansions were introduced and the effective-range approximation of cutting off the expansion after the second term was shown to be valid [39]. It is not obvious, however, that the effective-range approximation should be valid for ${}^7\text{Be} + p$ at the energies $E = 0.3\text{--}0.75$ MeV of the experiment [22], or that the effective range should be limited to small positive values.

Baye [32] obtained a formula for ${}^7\text{Be} + p$ for the dependence of s_1 on the mean scattering length $\bar{a}_s$ (essentially a_2)

$$s_1 \approx -1.81(1 + 0.0087\bar{a}_s \text{ fm}^{-1}) \text{ MeV}^{-1}. \quad (4)$$

It seems that this could be consistent with the values shown in Fig. 1 for $a = 4.0$ fm and for $a = 6.0$ fm, for reasonable values of b , but not for $a = 5.0$ fm. Baye neglected the effective range in obtaining Eq. (4). From Table 2, it is seen that $|r_{e2}|$ is very large for $a = 5.0$ fm (associated with the vanishing of a_2 for a near 5.0 fm); this could be the reason for the aberrant behaviour shown in Fig. 1 for $a = 5.0$ fm.

Fig. 1 also shows the MN and V2 points from the full GCM calculations of Ref. [29]; the values of s_1 are obtained by fitting the $S(E)$ values given in Table V of Ref. [29] for $E = 0.02$, 0.05 and 0.10 MeV, using the formula (11) of Ref. [5]. This gives $s_1 = -1.86 \text{ MeV}^{-1}$ for MN and -1.92 MeV^{-1} for V2. The value $s_1 = -1.97 \text{ MeV}^{-1}$ for DB is obtained from the values of a and c given in Table IV of Ref. [40]. We note that these values of s_1 would imply a positive value of $\bar{a}_s$ in Eq. (4), contrary to expectation.

The uncertainties associated with the use of the effective-range approximation in the analysis of the ${}^7\text{Be} + p$ elastic scattering data, and with the use of a potential model for relating the scattering-length and effective-range values in ${}^7\text{Li} + n$ and ${}^7\text{Be} + p$, may be avoided by using R -matrix formulae for describing the 1^- and 2^- contributions to the elastic scattering, with the R -matrix parameter values restricted by fitting ${}^7\text{Li} + n$ data.

For simplicity, we use the one-level, one-channel 2^- R -matrix approximation of Ref. [34], with parameters chosen for a given value of the channel radius a to fit the experimental data given in (2) (together with $S(1^+) = 0.48$, $S(2^+) = 0.87$). Then the fits to the ${}^7\text{Be} + p$ elastic-scattering data [22] involve nine adjustable parameters—the energy and two reduced-width amplitudes ($s = 1, 2$) of the 1^+ level of ${}^8\text{B}$, and six resolution parameters for the twenty angles. The best-fit parameter values are given in Table 3, group A. These correspond to group A of Table 2. Γ^0 is the observed width of the 1^+ level (found to be 31 ± 4 keV in Ref. [22]). The values in the “res” column give the range of values of the FWHM of the Gaussian distributions used for the resolution functions (the FWHM is 2.35 times the ΔE of Ref. [22]). The best fit is obtained for $a = 5.4$ fm (total $\chi^2 = 374$ for 542 data points and 9 adjustable parameters), and acceptable fits are obtained for a range of a values around this. These fits are therefore consistent with charge symmetry between the ${}^7\text{Li} + n$ and ${}^7\text{Be} + p$ systems.

In group B of Table 3, we give the best-fit parameter values when the energies and reduced widths of the lower 1^- level and of the 2^- level are allowed to vary (keeping fixed the separation of the 1^- levels and the reduced width of the upper 1^- level). Strong correlations between the 1^- and 2^- level parameters lead to fluctuations in the values given in Table 3. The best fit is for $a = 5.3$ fm, with $\chi^2 = 354$ (our best fit obtained using the effective-range description gives $\chi^2 = 357$, with $a_1 = 70.6$ fm, $r_{e1} = -0.07$ fm, $a_2 = -16.8$ fm and $r_{e2} = 7.74$ fm). The values of a_s and r_{es} corresponding to the values in group B of Table 3 are given in group B of Table 2. It may be noted that the values of a_s for ${}^7\text{Be} + p$ obtained by fitting the ${}^7\text{Be} + p$ elastic-scattering data [22] (group B of Table 2) are not very sensitive to the value of a (at least for $a \lesssim 5.5$ fm),

Table 3

Parameter values from fits to ${}^7\text{Be} + p$ elastic scattering data [22] (542 data points) in the one-level, one-channel 2^- approximation

Group	a (fm)	$E_1(1^-)$ (MeV)	$\gamma_1^2(1^-)$ (MeV)	$E_1(2^-)$ (MeV)	$\gamma_1^2(2^-)$ (MeV)	$E_r(1^+)$ (MeV)	$\gamma(1^+, s=1)$ (MeV $^{1/2}$)	$\gamma(1^+, s=2)$ (MeV $^{1/2}$)	$I^0(1^+)$ (keV)	res (keV)	χ^2
A	4.0	3.513 ^a	1.702 ^a	1.544 ^a	2.946 ^a	0.630	0.802	0.363	40	42–64	1713
	4.5	2.586 ^a	1.329 ^a	1.241 ^a	2.242 ^a	0.630	0.617	0.261	36	40–62	1164
	5.0	1.968 ^a	1.069 ^a	1.007 ^a	1.738 ^a	0.630	0.476	0.178	31	37–58	580
	5.3	1.701 ^a	0.952 ^a	0.896 ^a	1.510 ^a	0.630	0.407	0.133	28	35–54	383
	5.4	1.625 ^a	0.919 ^a	0.863 ^a	1.449 ^a	0.630	0.387	0.120	26	35–53	374
	5.5	1.555 ^a	0.888 ^a	0.832 ^a	1.381 ^a	0.630	0.366	0.104	25	34–51	410
	6.0	1.272 ^a	0.762 ^a	0.702 ^a	1.127 ^a	0.630	0.278	0.029	18	31–42	1612
	6.5	1.068 ^a	0.671 ^a	0.605 ^a	0.943 ^a	0.629	0.205	0.000	13	24–35	6210
B	4.0	42.9	1.75	1.34	2.79	0.633	0.687	0.109	28	35–50	436
	4.5	9.10	1.05	1.27	2.39	0.632	0.557	0.106	28	35–50	399
	5.0	6.84	0.597	0.638	1.92	0.633	0.497	0.118	32	31–51	361
	5.3	2.24	0.249	0.710	1.86	0.633	0.448	0.121	32	30–51	354
	5.4	1.83	0.201	0.473	1.65	0.633	0.440	0.126	33	29–52	357
	5.5	1.49	0.260	0.441	1.54	0.632	0.423	0.128	33	29–52	361
	6.0	1.53	0.813	0.239	1.13	0.630	0.353	0.140	32	29–55	439
	6.5	0.660	0.706	0.301	1.08	0.629	0.305	0.152	33	27–59	710

^a Fixed value.

and show some similarity to the values obtained in Ref. [22]. Also all the fits in Table 3 give a channel-spin ratio for the 1^+ level that is appreciably less than one, as found in Ref. [22].

This is not intended to show that charge symmetry necessarily holds for the ${}^7\text{Li} + n$ and ${}^7\text{Be} + p$ systems. It does suggest, however, that the argument that charge symmetry does not hold may not be valid.

5. Conclusion

The energy dependence of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ S factor predicted by the DB cluster model is in better agreement with experiment than is that from other models. This is attributed to DBs low value of the slope parameter b . We here suggest that DBs b is low because their value of $S(0)$ is high, implying a large value of the spectroscopic factor $S(2^+)$. This changes the energy dependence of the 2^- contribution to $S(E)$, provided that ${}^7\text{Li} + n$ data for thermal neutrons are fitted. The suggestion that use of ${}^7\text{Li} + n$ data is not justified due to a breakdown of charge symmetry is discussed and seems not to be convincing.

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